

Addressing Foundational Dysfunctions on the Nodes of the Matrix

Introduction



ROBERT LUBY, MD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Robert Luby, MD, disclosed no relevant financial relationships with any ineligible companies.

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Functional Medicine Certification Competencies

II. PATIENT EVALUATION

a. EVALUATION OF PATIENT DOMAINS OF FUNCTION

Identify evidence of disturbances in function and ability to adapt to changing environments and allostatic load in the patient's past and current clinical indicators. These include symptoms, physical signs, health conditions, biometric data/test results, or overt signs of pathology in the following domains:

- i. Production of energy and mitigation or modulation of the impact of oxidative stress and inflammation.
- ii. Assimilation of nutrients, other substances (e.g., medications, hormones, toxicants), environmental factors, experiences, information (e.g., mental, emotional, spiritual) into the body and mind.
- iii. Transport of compounds, substances, and energy among cells, tissues, organs, and body systems.
- iv. Production and balance of compounds (e.g., hormones, neurotransmitters) that mediate biochemical signals and information.
- v. Maintenance and optimization of structural integrity of the body from microscopic to macroscopic (e.g., cellular membrane to musculoskeletal system).
- vi. Immune regulation, microbiome balance, and repair of inflamed or damaged tissues.
- vii. Biotransformation and elimination of toxins, toxicants, and waste products.

Functional Medicine Certification Competencies

II. PATIENT EVALUATION

b. EVALUATION AND SYNTHESIS OF ROOT CAUSES

Elicit and understand personalized factors that enable the diagnosis of upstream disturbances in function and disease and understand when to refer for further evaluation to:

- i. Identify the latent root causes of current disturbances in function, inherited or acquired, that were predispositions (antecedents) to the patient's current health conditions. These latent root causes include genetics, exposures, significant past health events, behaviors/habits, circumstances, and other factors (e.g., diet, lifestyle, mental/emotional/spiritual health, social determinants of health [SDOH], occupational hazards, etc.).
- ii. Identify the rapidly acting root causes of disturbances in function throughout the course of the patient's life that were, or are, immediate activators/promoters (triggers) of their health conditions. These activating root causes include exposures, significant past and current life events, behaviors/habits, circumstances, and other factors (e.g., diet, lifestyle, mental/emotional/spiritual health, social determinants of health [SDOH], occupational hazards, etc.).
- iii. Identify the current, ongoing, or frequently occurring factors that serve as perpetuating root causes (mediators) of disturbances in function and whose presence prevents improvement of function. These perpetuating root causes include exposures, significant life events, behaviors/habits, circumstances, and other factors (e.g., diet, lifestyle, mental/emotional/spiritual health, social determinants of health [SDOH], occupational hazards, etc.).
- iv. Organize and synthesize information from the patient evaluation to construct a timeline of life events and factors that serve(d) as personalized root causes of disturbances in function.

Functional Medicine Certification Competencies

b. EVALUATION AND SYNTHESIS OF ROOT CAUSES

v. Synthesize, integrate, and correlate the mechanisms of action of the clinical indicators and personalized root causes identified in the patient evaluation to identify, organize, and prioritize patterns of disturbance in function and clinical imbalances in the following domains:

1. Production of energy and mitigation or modulation of the impact of oxidative stress & inflammation.
2. Assimilation of nutrients, other substances (e.g., medications, hormones, toxicants), environmental factors, experiences, information (e.g., mental, emotional, spiritual) into the body and mind.
3. Transport of compounds, substances, and energy among cells, tissues, organs, and body systems.
4. Production & balance of compounds (e.g., hormones, neurotransmitters) that mediate biochemical signals and information.
5. Maintenance and optimization of structural integrity of the body from microscopic to macroscopic (e.g., cellular membrane to musculoskeletal system).
6. Immune regulation, microbiome balance, and repair of inflamed or damaged tissues.
7. Biotransformation and elimination of toxins, toxicants, and waste products.

Functional Medicine Certification Competencies

II. PATIENT EVALUATION

d. LABORATORY AND OTHER EVALUATIVE TESTING

- i. Determine if further quantitative data is necessary to provide additional predictive, prognostic, and/or diagnostic specificity after completion of the qualitative patient evaluation.
- ii. In these scenarios, perform, interpret, or refer for evaluative tests using analytes and biomarkers capable of providing clinical evidence of disturbances in physiological, structural, mental, or emotional function, and that may improve predictive, prognostic, and/or diagnostic specificity.

Functional Medicine Certification Competencies

III. THERAPEUTIC INTERVENTION

- a) Evaluate patient readiness to engage in dietary, lifestyle, and behavioral changes prior to initiating therapeutic interventions; prioritize these as initial interventional modalities.
- b) According to their ability to favorably modulate function, correct clinical imbalances, and/or address each patient's personalized root causes in alignment with the patient's goals, values, preferences, sociocultural context, and what gives them meaning and purpose:
 - i. Identify, prioritize, select, and initiate the following therapeutic interventions when clinically indicated:
 1. Dietary and other lifestyle strategies.
 2. Behavioral change strategies.
 3. Pharmaceuticals, nutraceuticals, botanicals, and other supplements.
 4. Mind-body and other modalities.
 - ii. Monitor changes in function in response to therapeutic interventions and adjust regimen accordingly.
 - iii. Refer and/or collaborate with additional healthcare practitioners, as needed or required, to align with the patient's goals and desired outcomes.

Clinical Guidelines

UK: Management of Pancreatic Exocrine Insufficiency
<https://bmjopengastro.bmj.com/content/8/1/e000643>

NIH: Malabsorption Syndromes: “StatPearls”
<https://www.ncbi.nlm.nih.gov/books/NBK553106/>

Rome Foundation: Rome IV Diagnostic Criteria
<https://theromefoundation.org/rome-iv/rome-iv-criteria/>

Clinical Guidelines

NIH: Intestinal Permeability Disturbances – Causes, Diseases, and Therapy

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11438725/#Bib1>

Australia Society of Intestinal Permeability: Management of Increased IP

<https://static1.squarespace.com/static/638330a7b6aad91b6a65564a/t/63c8b5cf882fd2748b9fc6e7/1674098139579/IP+Guideline+By+Dr+Brad+Leech.pdf>

Clinical Guidelines

AGA: Fecal Microbiota-Based Therapies for Select GI Diseases

[https://www.gastrojournal.org/article/S0016-5085\(24\)00041-6/fulltext#:~:text=The%20guideline%20panel%20made%207,the%20context%20of%20clinical%20trials.](https://www.gastrojournal.org/article/S0016-5085(24)00041-6/fulltext#:~:text=The%20guideline%20panel%20made%207,the%20context%20of%20clinical%20trials.)

NIH: Gut Microbiome Health and Dysbiosis: A Clinical Primer

[https://pmc.ncbi.nlm.nih.gov/articles/PMC9827978/#:~:text=One%20proposed%20dysbiosis%20index%20is%20a%20ratio,the%20total%20number%20of%20overall%20bacterial%20strains.&text=Clinicians%20can%20promote%20a%20diet%20rich%20in,microbiome%20\(e.g.%2C%20as%20a%20source%20of%20SCFA\).](https://pmc.ncbi.nlm.nih.gov/articles/PMC9827978/#:~:text=One%20proposed%20dysbiosis%20index%20is%20a%20ratio,the%20total%20number%20of%20overall%20bacterial%20strains.&text=Clinicians%20can%20promote%20a%20diet%20rich%20in,microbiome%20(e.g.%2C%20as%20a%20source%20of%20SCFA).)

Clinical Guidelines

NIH: CPG for Inflammatory Disease

<https://pubmed.ncbi.nlm.nih.gov/29429045/#:~:text=Although%20etiology%20is%20still%20uncertain,Inflammatory%20bowel%20disease;%20Ulcerative%20colitis.>

AGA: Diet and Nutritional Therapies in IBD

[https://www.gastrojournal.org/article/S0016-5085\(23\)05597-X/fulltext#:~:text=All%20patients%20with%20IBD%20warrant,Best%20Practice%20Advice%2012](https://www.gastrojournal.org/article/S0016-5085(23)05597-X/fulltext#:~:text=All%20patients%20with%20IBD%20warrant,Best%20Practice%20Advice%2012)

Clinical Guidelines

Rome Foundation: Rome V Criteria – DGBI (available 2026)

<https://theromefoundation.org/rome-iv/rome-v/>

AGA: Clinical Practice Update on GI Pain in DGBI

<https://www.sciencedirect.com/science/article/pii/S1542356521007175>

IDSA: CPG for Infectious Diarrhea

<https://www.idsociety.org/practice-guideline/infectious-diarrhea/>

ARUP: Inflammatory Markers

<https://arupconsult.com/content/inflammatory-markers>

Clinical Guidelines

British Society of Gastroenterology: Consensus Guidelines on IBD

<https://pubmed.ncbi.nlm.nih.gov/31562236/#:~:text=Comprehensive%20up%2Dto%2Ddate%20guidance,;%20virtual%20clinic;%20vitamin%20D.>

Japanese Society of Gastroenterology: CPG on IBD

<https://link.springer.com/article/10.1007/s00535-021-01784-1>

Clinical Guidelines

Japan Endocrine Society: Adrenal Insufficiency

https://www.jstage.jst.go.jp/article/endocrj/63/9/63_EJ16-0242/_pdf

Endocrine Society: Primary Adrenal Insufficiency CPG

<https://www.endocrine.org/clinical-practice-guidelines/primary-adrenal-insufficiency>

Endocrine Society: Cushing's Syndrome CPG

<https://academic.oup.com/jcem/article/100/8/2807/2836065>

WHO: Guidelines for Acute Stress, PTSD, Bereavement

<https://www.who.int/publications/i/item/9789241505406>

Clinical Guidelines

AHA: Lifestyle and Cardio Health (Pillars)

<https://www.ahajournals.org/doi/full/10.1161/CIR.0000000000001018>

AHA: Primary Prevention of Cardiovascular Disease

<https://www.ahajournals.org/doi/10.1161/cir.0000000000000678#:~:text=For%20adults%20with%20overweight%20and,a%20clinician%E2%80%93patient%20risk%20discussion.>

Clinical Guidelines

ESC: Management of Cardiovascular Disease in Patients With Diabetes

<https://www.escardio.org/Guidelines/Clinical-Practice-Guidelines/CVD-and-Diabetes-Guidelines#:~:text=Based%20on%20evidence%20from%20large,learning%20or%20machine%20learning%20models.>

Poland Medical Societies (Multiple): Metabolic Syndrome Management Guidelines

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9479724/>

AHA/NHLBI: Diagnosis and Management of Metabolic Syndrome (2005)

<https://www.ahajournals.org/doi/10.1161/circulationaha.105.169404>

Clinical Guidelines

American Academy of Clinical Toxicology: Position Statements

<https://www.clintox.org/resources/position-statements>

Mitochondrial Medicine Society: Diagnosis and Management of Mitochondrial Disease

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5000852/#:~:text=There%20are%20no%20published%20consensus,of%20mitochondrial%20disease%20is%20derived.&text=There%20are%20also%20inconsistencies%20in,of%20this%20consensus%20development%20endeavor>.

ACGS (UK): Molecular Diagnosis of Mitochondrial Disease

https://www.acgs.uk.com/media/11935/bpg-for-the-molecular-diagnosis-of-mitochondrial-disease_ratified-november-2020.pdf

Learning Objectives

At the conclusion of this segment, successful participants will be able to:

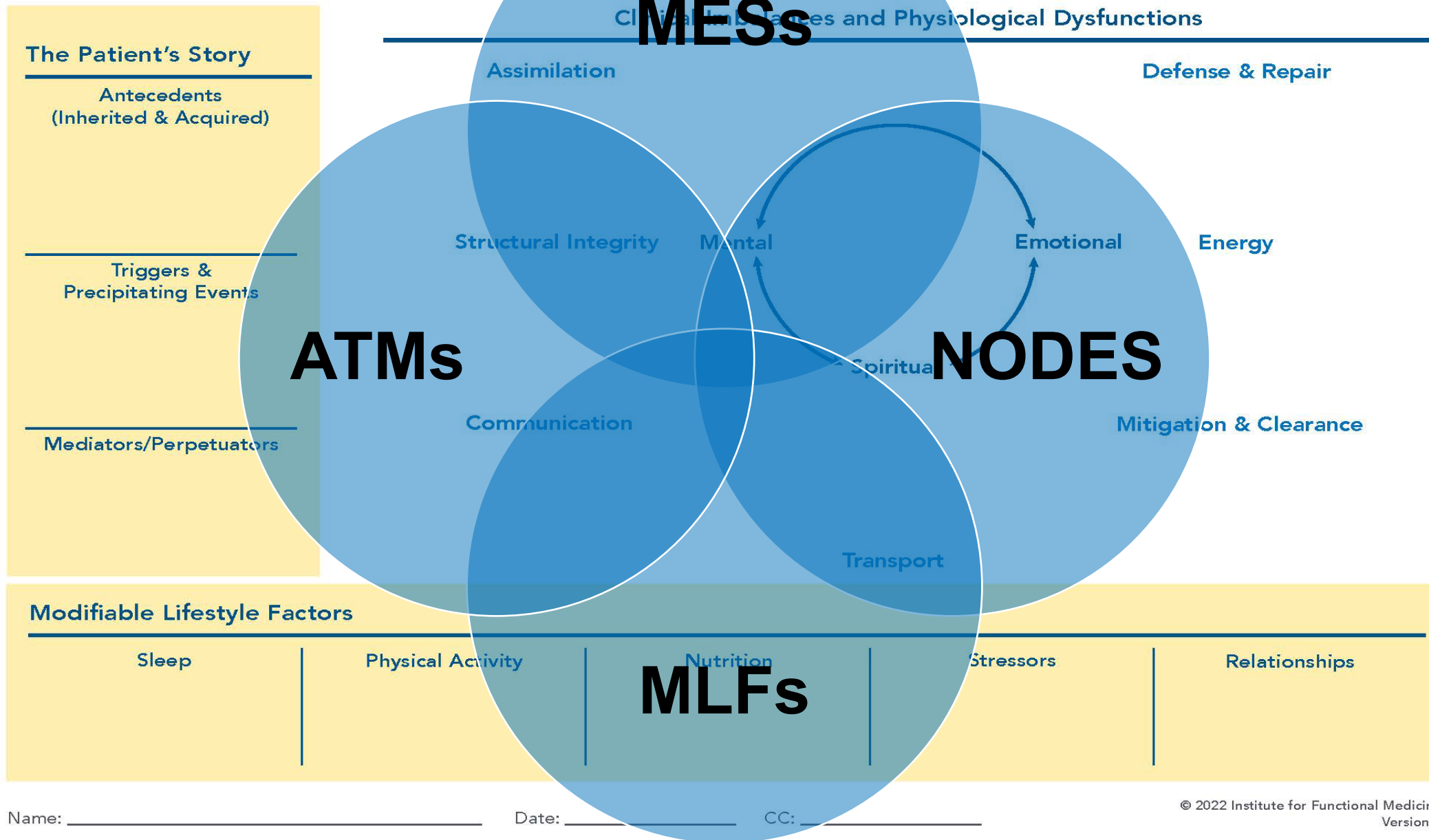
1. List and explain what data is populated in the different sections of the functional medicine clinical Matrix.
2. Explain how an accurately populated Matrix enables participants to identify patterns of physiological dysfunction and clinical imbalances.
3. Explain how the patterns of dysfunction inform the prioritization of personalized interventions that address the root causes of the patient's current concerns.

Clinical Focus

- Relationships among the various sections and the seven nodes of the matrix.



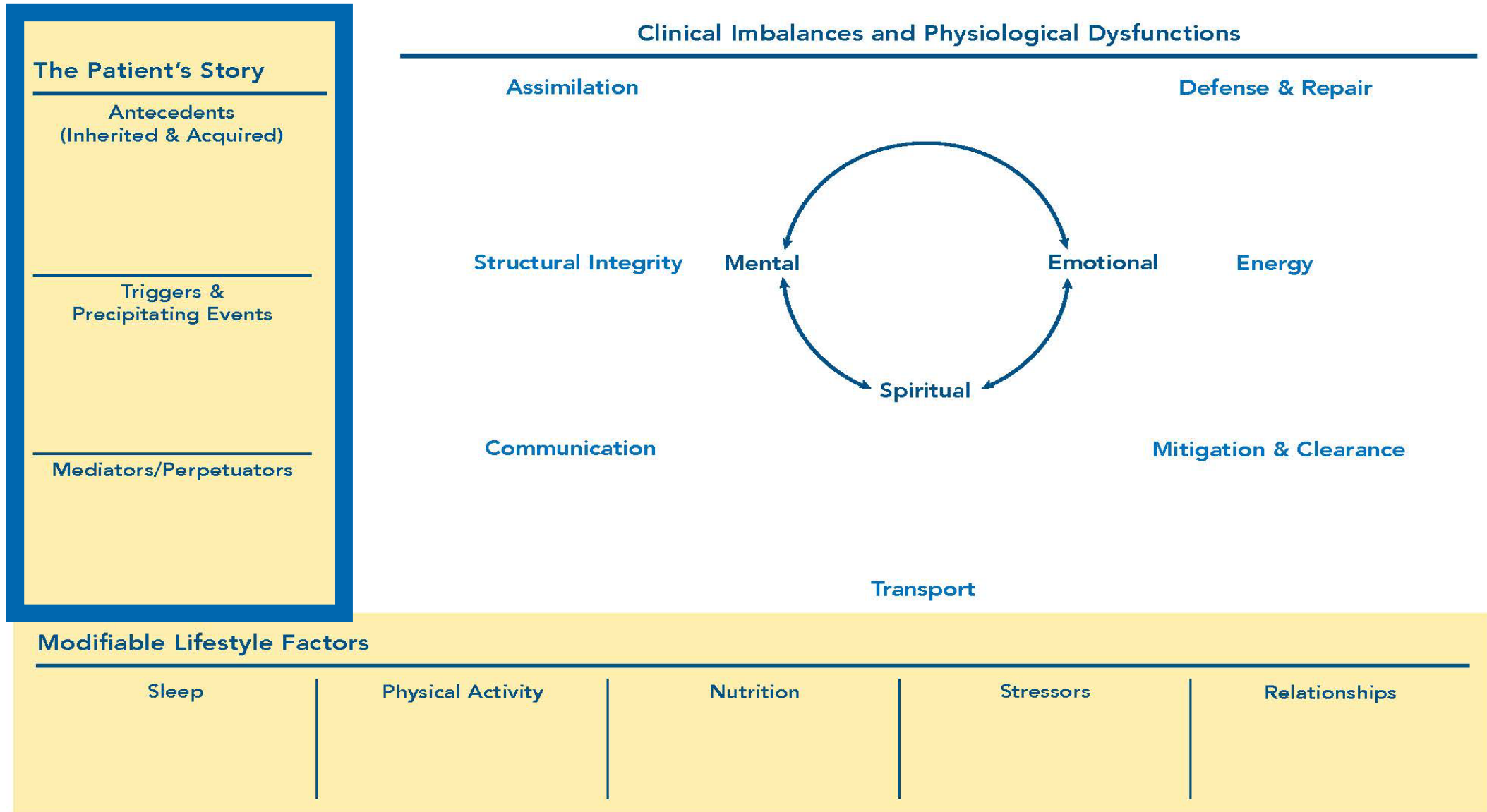
Functional Medicine Matrix



Long Description: Functional Medicine Matrix

- The Functional Medicine Matrix includes the following:
- The Patient's Story:
 - Antecedents – inherited and acquired
 - Triggers and precipitating events
 - Mediators/perpetuators
- Modifiable Lifestyle Factors: sleep, physical activity, nutrition, stressors, relationships
- Clinical Imbalances and Physiological Dysfunctions: assimilation, defense and repair, energy, mitigation and clearance, transport, communication, structural integrity
- Mental, emotional, and spiritual
- This image of an empty matrix is overlaid with 4 overlapping circles, each representing M E S's, Nodes, M L F's, and A T M's

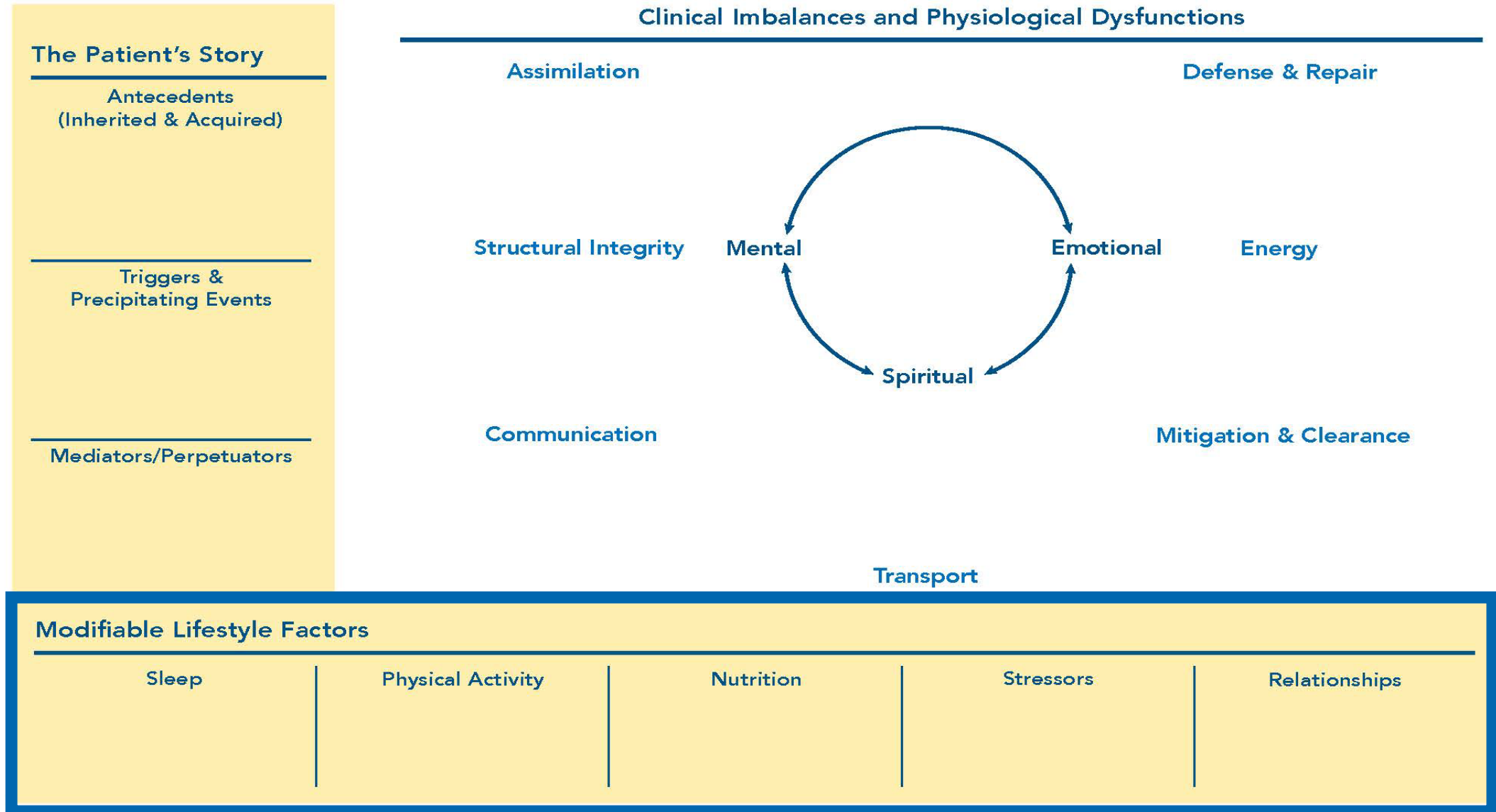
Functional Medicine Matrix



Long Description: Functional Medicine Matrix

- The Functional Medicine Matrix with an emphasis on the Patient's Story
- Antecedents – inherited and acquired
- Triggers and precipitating events
- Mediators/perpetuators

Functional Medicine Matrix



Long Description: Functional Medicine Matrix

- The Functional Medicine Matrix with an emphasis on Modifiable Lifestyle Factors including sleep, physical activity, nutrition, stressors, and relationships

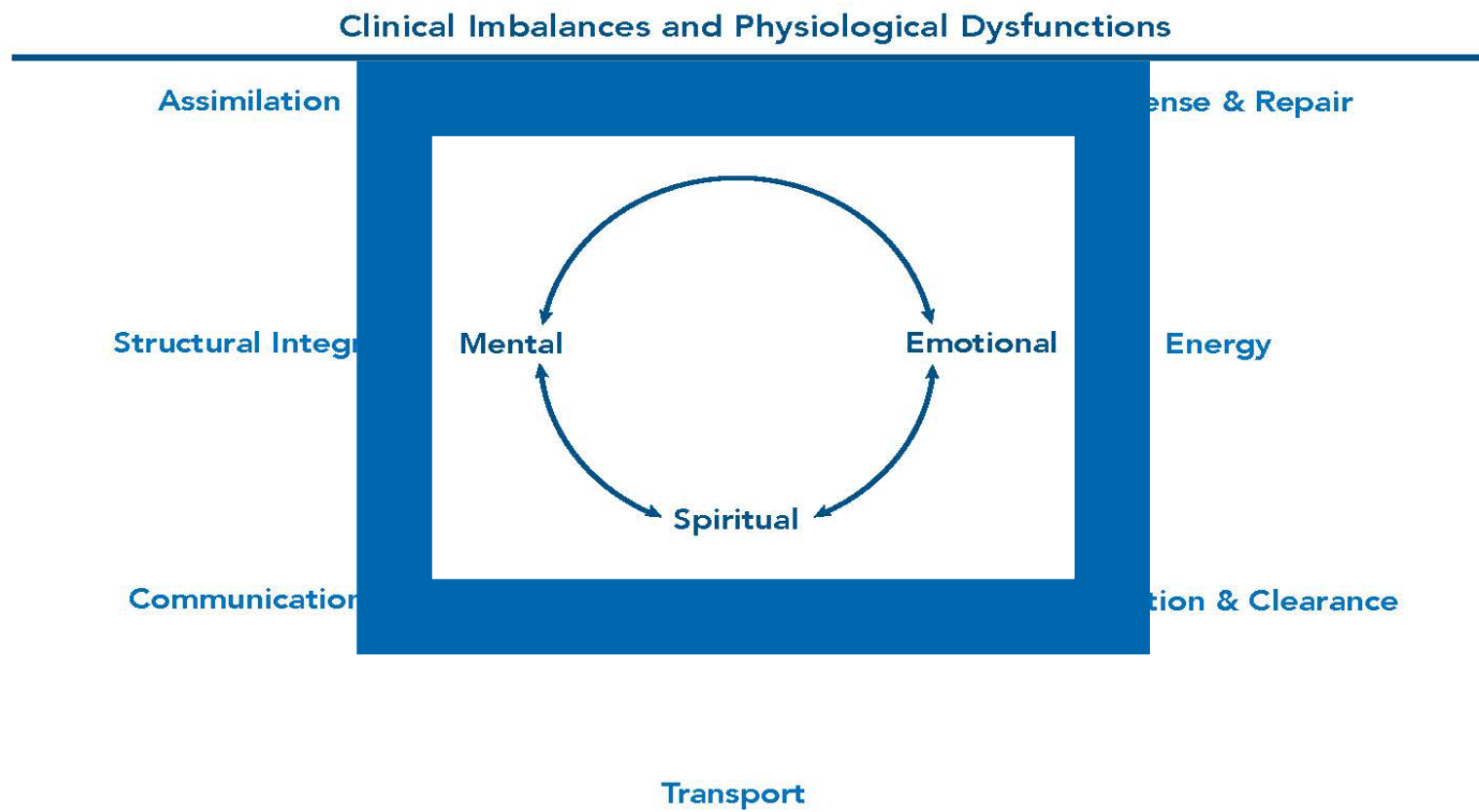
Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators



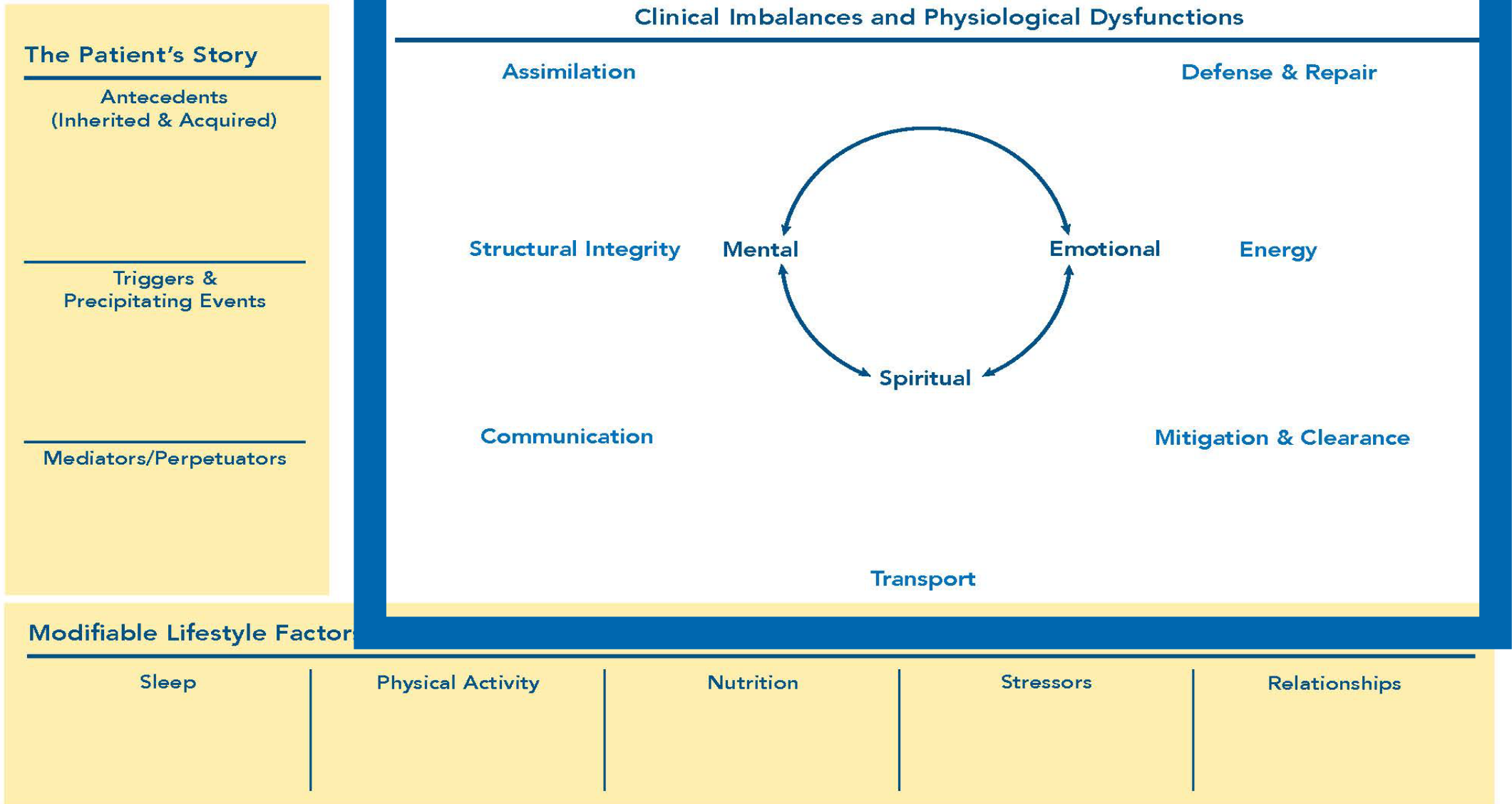
Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Long Description: Functional Medicine Matrix

- The Functional Medicine Matrix with an emphasis on the mental, emotional, and spiritual nodes.

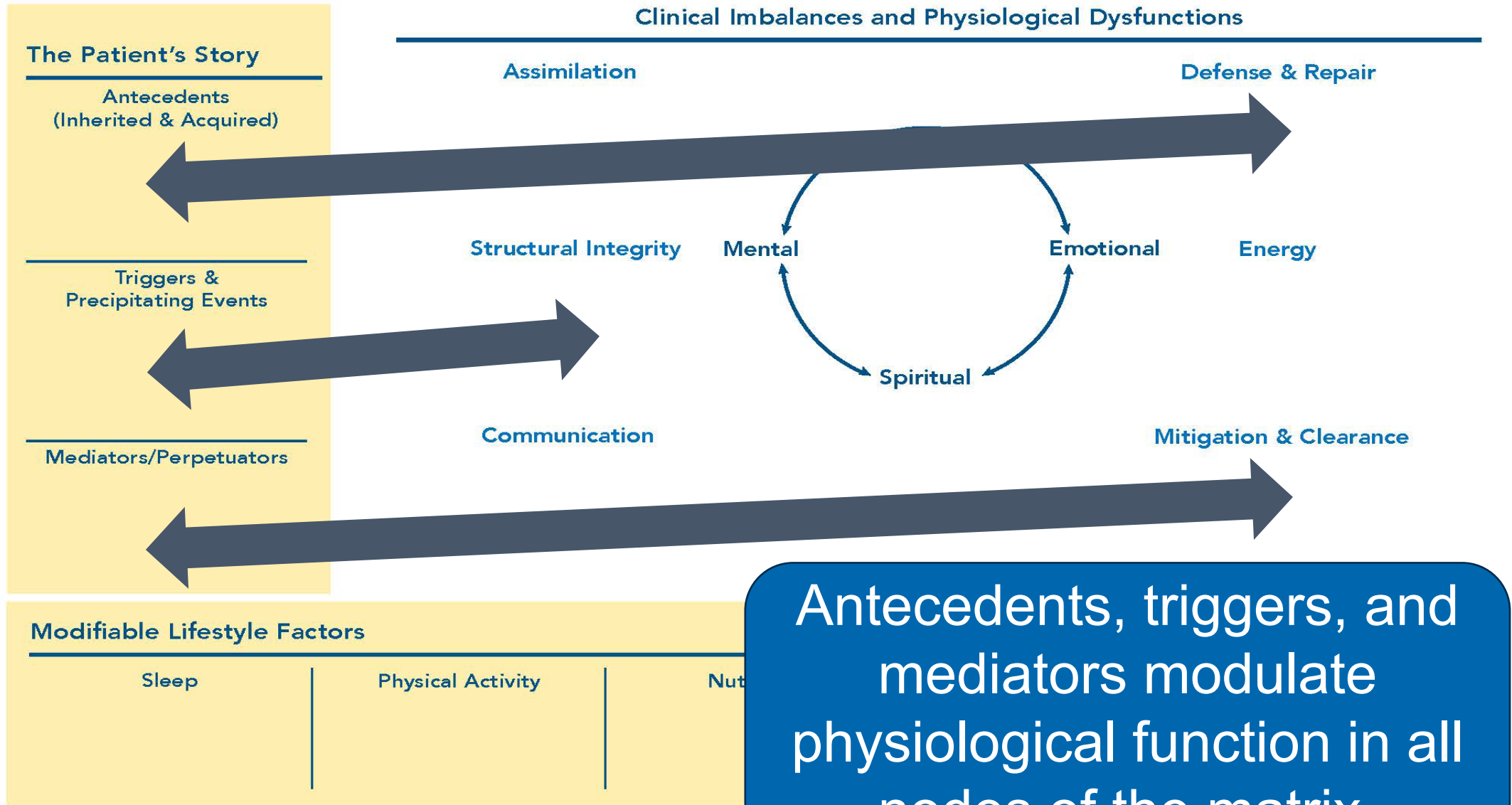
Functional Medicine Matrix



Long Description: Functional Medicine Matrix

- The Functional Medicine Matrix with an emphasis on the clinical imbalances and physiological dysfunctions including assimilation, defense and repair, energy, mitigation and clearance, transport, communication, structural integrity, and M E S.

Functional Medicine Matrix

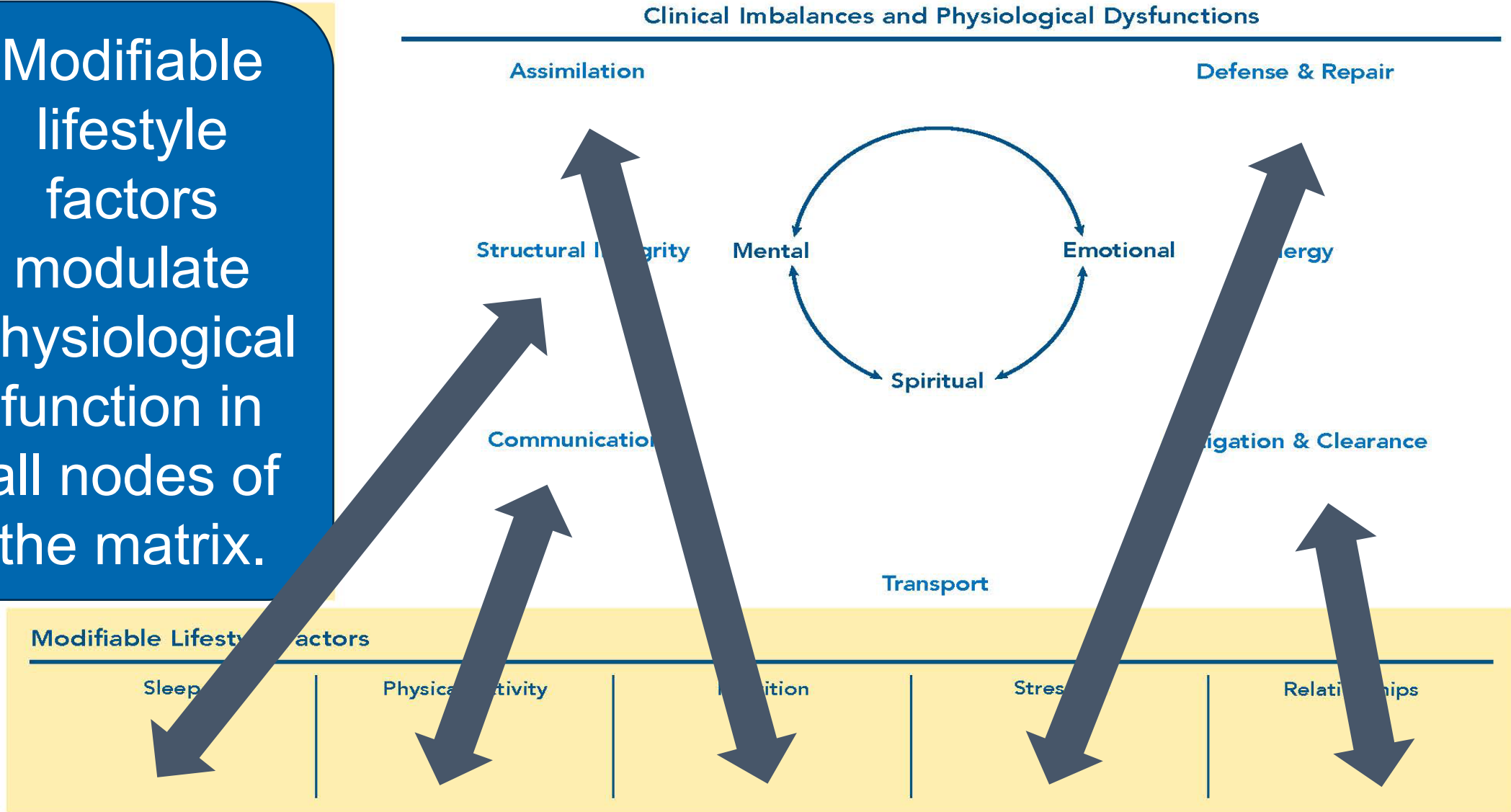


Long Description: Functional Medicine Matrix

- The Functional Medicine Matrix demonstrating how antecedents, triggers, and mediators modulate physiological function in all nodes of the matrix.

Functional Medicine Matrix

Modifiable lifestyle factors modulate physiological function in all nodes of the matrix.



Long Description: Functional Medicine Matrix

- The Functional Medicine Matrix demonstrating how modifiable lifestyle factors modulate physiological function in all nodes of the matrix.

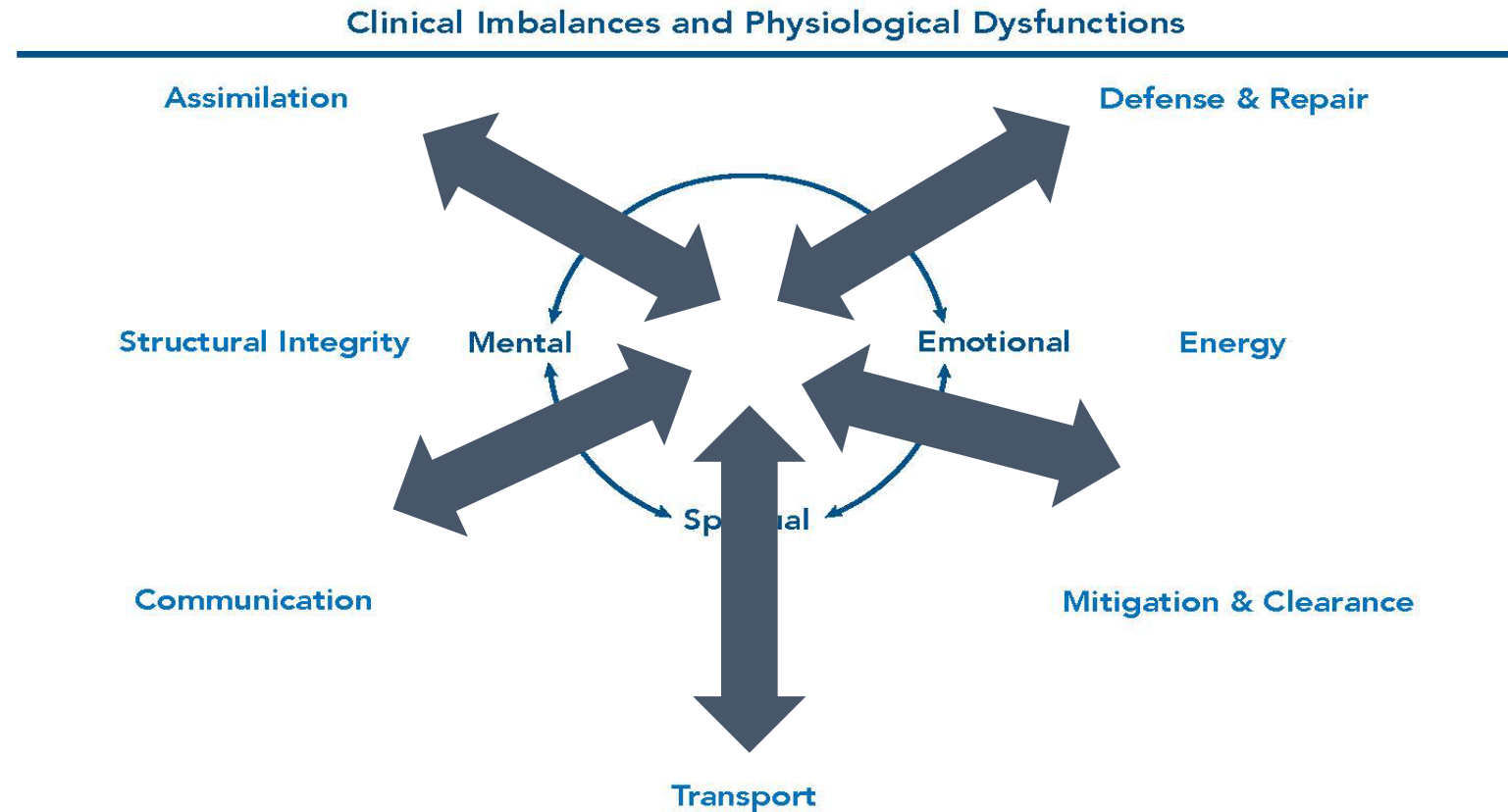
Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators



Mental, emotional, and spiritual factors modulate physiological function in all nodes of the matrix.

Stressors	Relationships
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Long Description: Functional Medicine Matrix

- The Functional Medicine Matrix demonstrating how mental, emotional, and spiritual factors modulate physiological function in all nodes of the matrix.

Clinical Focus

- Metaphor: The nodes of the matrix as the primary movers of physiological function.



Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Assimilation
(e.g., Digestion,
Absorption,
Microbiota/GI,
Respiration, Skin,
Mucous Membranes)

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on assimilation – digestion, absorption, microbiota/ G I, respiration, skin, mucous membranes

Functional Medicine Matrix

Assimilation

External → internal
movement of
substances across
barriers and
membranes.

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Name: _____

Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on assimilation: external to internal movement of substances across barriers and membranes.

Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

**Mitigation &
Clearance**
(Toxicant Exposure,
Avoidance, Mitigation,
Biotransformation,
Elimination, Clearance)

Name: _____

Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Mitigation and clearance: toxicant exposure, avoidance, mitigation, biotransformation, elimination, clearance

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Factors/Perpetuators

Mitigation & Clearance

Internal → external
movement of
substances.

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Name: _____

Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Mitigation and clearance: internal to external movement of substances

Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Internal to external movement of substances

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport
(Vasculature, Lymph,
Intracellular,
Transmembrane)

Transport

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Name: _____

Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Transport: vasculature, lymph, intracellular, transmembrane

Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Transport

Internal → internal
movement of
substances and agents
across barriers and
membranes.

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Transport: internal to internal movement of substances and agents across barriers and membranes

Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Communication
(Endocrine,
Neurotransmitters,
Hormones, Immune
Messengers)

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Name: _____

Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Communication: endocrine, neurotransmitters, hormones, immune, messengers

Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Communication

Information mediated by molecules and other agents if/when successfully transported across barriers and membranes.

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Communication: information mediated by molecules and other agents if/when successfully transported across barriers and membranes

Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Structural Integrity
(Subcellular Membranes, Epithelium, Endothelium, Fascia, Musculoskeletal)

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Structural Integrity: subcellular membranes, epithelium, endothelium, fascia, musculoskeletal

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Structural Integrity

The barriers, membranes, and structures across which substances are transported.

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Structural Integrity: the barriers, membranes, and structures across which substances are transported

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

**Defense &
Repair**
(Immune,
Inflammation,
Autoimmunity,
Infections)

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Name: _____

Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Defense and Repair: immune, inflammation, autoimmunity, infections

Functional Medicine Matrix

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Defense & Repair

That which repairs the structures and defends against the “breachers of barriers.”

Name: _____

Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Defense and Repair: that which repairs the structures and defends against the “breachers of barriers”

Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Energy
(Energy Regulation,
Mitochondrial
Function)

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Name: _____

Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Energy: energy regulation, mitochondrial function

Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Energy

Generates the energy
that fuels the movement
of all the other functions.

Name: _____

Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on Energy: generates the energy that fuels the movement of all the other functions

Clinical Focus

- The differential diagnosis of dysfunction and the matrix.



The Functional Medicine Clinical Method

How do you **ORGANIZE** what you have **GATHERED**?



THE INSTITUTE FOR
FUNCTIONAL
MEDICINE®

GOTOIT

The sequence of steps used in the practice of functional medicine:

GOTOIT Step and Definition	IFM Tools (examples)	Clinical Actions
G – GATHER oneself with a brief personal practice prior to prepare for each patient encounter. Take a detailed functional medicine history that includes the essential health events and features of a patient's life documented on the IFM Timeline in the process of organizing the patient's data.	• Cultivating Self Awareness and Mindfulness • Adult Intake Form • Medical Symptom Questionnaire (MSQ) • PROMIS Questionnaires • Diet, Nutrition, and Lifestyle Journal • Functional Medicine Tree • Functional Medicine Timeline	• Use the Cultivating Self-Awareness and Mindfulness document to help gather yourself before every patient encounter. • The Adult Intake Form is used to gather initial intake information. Whereas, the MSQ, PROMIS Questionnaires, and Diet, Nutrition, and Lifestyle Journal help gather initial symptom, function, and dietary information. • Map root causes (antecedents, triggers, mediators (ATMs)) and effects (clinical findings, current concerns) of physiological dysfunction on the Functional Medicine Timeline.
O – ORGANIZE the data gathered in the previous step and document them on the IFM Timeline and Matrix to identify the personalized root causes of current concerns, dysfunctions, and conditions, and to begin to prioritize the order (sequence) of interventions.	• Functional Medicine Tree • Functional Medicine Timeline • Functional Medicine Matrix • Adult Intake Form • Medical Symptom Questionnaire (MSQ) • PROMIS Questionnaires • Diet, Nutrition, and Lifestyle Journal	• Using the Functional Medicine Matrix, populate ATMs, clinical findings, and current concerns from the Timeline. Add modifiable lifestyle factors (MLFs) and mental, emotional, spiritual factors (MES) to support the identification of the most important areas of physiological dysfunction.
T – TELL the life story back to the patient to ensure accuracy and a shared understanding. This dialogue, focused on the Timeline, highlights how factors elicited in the Gather step may have contributed to current or future clinical findings, concerns, health status and trajectory. The patient is asked to join in correcting the story, engendering a context of true partnership.	• Functional Medicine Matrix • Functional Medicine Timeline • Functional Medicine Tree	• The Functional Medicine Timeline is used to re-tell the patient's story while correlating prior events and root causes with current concerns or conditions. Ask the patient to join in for corrections or additional context. • Use the Functional Medicine Tree to explain how the root causes have contributed to the patient's condition.

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GOTOIT Step and Definition	IFM Tools (examples)	Clinical Actions
O – ORDER and collaboratively prioritize the most feasible and effective interventions by taking antecedents, triggers, mediators, and patient contextual factors into account.	• Functional Medicine Matrix • Readiness for Change • Goal Setting for Behavior Change • Integrating Guided Imagery and Visualization to Achieve Personal Health Goals	• Using the Functional Medicine Matrix, co-prioritize the areas of physiological dysfunction. • Assess if the patient is ready and committed to making changes with the Readiness for Change document. Use the Goal Setting for Behavior Change document to determine the patient's goals, values, preferences, and priorities. Then set SMART goals for interventions.
I – INITIATE Initiate interventions based on collaborative prioritization: 1) Evaluative, diagnostic, and prognostic testing. 2) Referral to another member of the collaborative care team for further assessment, intervention, or support. 3) Therapeutic interventions.	• Functional Medicine Tree • Functional Medicine Matrix • Functional Medicine Timeline • Goal Setting for Behavior Change • Functional Medicine Prescription and Lifestyle Plan	• The Functional Medicine Prescription and Lifestyle Plan helps layout the therapeutic plan. This may include removing or reducing triggers, modifying lifestyle and dietary mediators, addressing gut health and function, implementing other therapeutic interventions, ordering appropriate laboratory tests, or referring as needed.
T – TRACK and monitor the effectiveness of, and adherence to, the initiated interventions as well as interval life events and changes in the patient's circumstances.	• Medical Symptoms Questionnaire (MSQ) • PROMIS Questionnaires • Diet, Nutrition, and Lifestyle Journal	• Use tools to track progress by comparing scores at subsequent visits. The MSQ helps track symptom trajectory, the PROMIS Questionnaires help track overall well-being and quality of life, and the Diet, Nutrition, and Lifestyle Journal helps track adherence to regimen. • There are many other tools and questionnaires that can be helpful for tracking progress.

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Long Description: The Functional Medicine Clinical Method

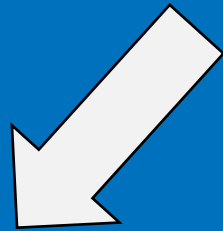
- Toolkit item – G O T O I T – the sequence of steps used in the practice of functional medicine

What disease does this patient have?

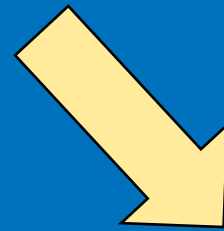
Why does *this* patient have
this disease?

What physiological dysfunctions contribute to this patient's condition, and how did those dysfunctions arise?

Differential Diagnosis of:



Disease



Dysfunction

**20th
Century**

Take History

**Effects
Symptoms**

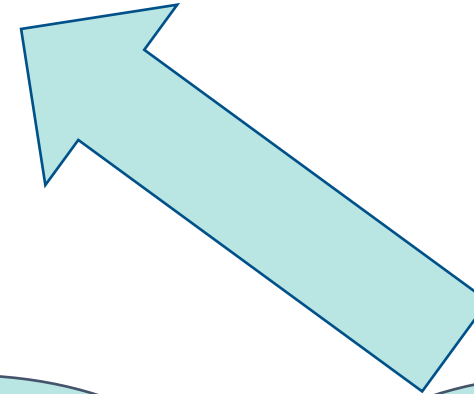
**Diagnose
and **Reverse**
the Disease**

**21st
Century**

**Take
History**

**Effects
Symptoms**

**Diagnose and
Manage the
Disease**



Long Description: No Title

- The flow path of 20th century medicine: take history, determine effects and symptoms, diagnose and reverse the disease
- The flow path of 21st century medicine: take history, determine effects and symptoms, diagnose and manage the disease

Clinical Focus

- Foundational physiological dysfunctions that will be addressed in the upcoming presentations of this course.



Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Substances

Relationships

Clinical Imbalances and Physiological Dysfunctions

Assimilation

DIGIN
5R

Structural Integrity

Intestinal
Permeability

Communication

Adrenal
Dysregulation

Defense & Repair

Chronic
Inflammation

Energy

Mitochondrial
Dysfunction

Mitigation & Clearance

Biotrans-
formation

Transport

Insulin
Resistance

Mental

Emotional

Spiritual

Long Description: The Functional Medicine Matrix

- Examples of nodes on the functional medicine matrix:
- Assimilation: D I G I N 5 R
- Defense and Repair: chronic inflammation
- Energy: mitochondrial dysfunction
- Mitigation and Clearance: biotransformation
- Transport: insulin resistance
- Communication: adrenal dysregulation
- Structural Integrity: intestinal permeability

Functional Medicine Matrix

Assimilation
(e.g., Digestion,
Absorption,
Microbiota/GI,
Respiration, Skin,
Mucous Membranes)

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Name: _____

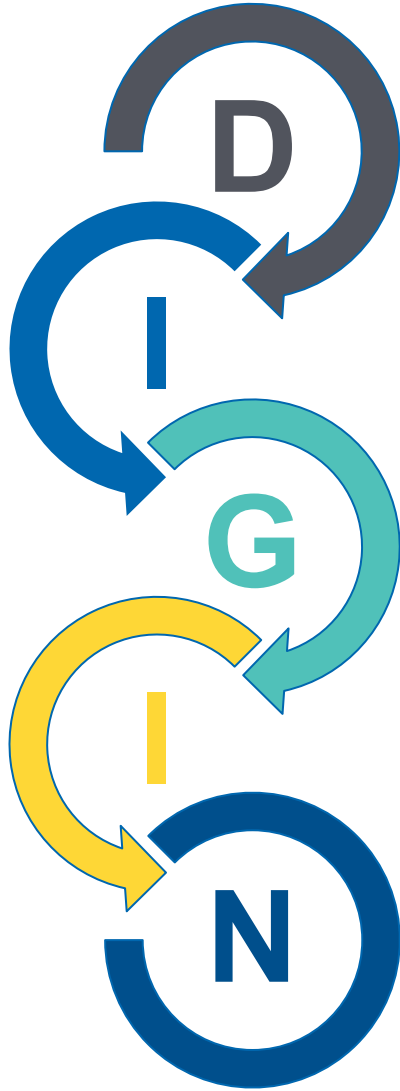
Date: _____

CC: _____

Long Description: Functional Medicine Matrix

- The functional medicine matrix with an emphasis on assimilation – digestion, absorption, microbiota/ G I, respiration, skin, mucous membranes

The DIGIN Mnemonic of Gut Dysfunctions



Digestion/Absorption

Intestinal Permeability

Gut Microbiota/Dysbiosis

Inflammation/Immune

Nervous System

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM

Long Description: The DIGIN Mnemonic of Gut Dysfunctions

D: Digestion/absorption

I: Intestinal permeability

G: Gut microbiota/dysbiosis

I: Inflammation/immune

N: Nervous system

Addressing Foundational Dysfunctions on the Nodes of the Matrix

Introduction

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE